Part 1: Theoretical Understanding

1. Short Answer Questions

Q1: Primary Differences Between TensorFlow and PyTorch

The main difference lies in their design philosophy regarding the **computational graph**:

spaCy Feature	**Basic Python Operation**	**Enhancement over Basic String Ops**
Tokenization	`.split(' ')` (splits only by whitespace)	Splits text intelligently, recognizing complex boundaries like punctuation, contractions ("don't"), and abbreviations ("U.S.A."), retaining linguistic integrity.
Lemmatization	Regex to strip suffixes (naive)	Finds the true **base form (lemma)** of a word considering its context and Part-of-Speech (POS) tag (e.g., converts "running" or "ran" to "run").
Named Entity Recognition (NER)	Regex patterns for capitalization/dates	Identifies and labels real-world objects in text (e.g., **PERSON**, **ORG**, **GPE**, **MONEY**) using a trained statistical model, enabling true **information extraction**.

2. Comparative Analysis: Scikit-learn and TensorFlow

Feature	**Scikit-learn (Sklearn)**	**TensorFlow (TF)**
Target Applications	**Classical/Traditional ML:** Classification (SVM, Decision Trees), Regression (Linear, Logistic), Clustering (K-Means), Dimensionality Reduction (PCA).	**Deep Learning (DL):** Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Transformers. Used for Computer Vision, advanced NLP, and Reinforcement Learning.
Ease of Use for Beginners	**Very High.** Features a high-level, consistent API based on the `Estimator` pattern (`.fit()`,	**Medium/Steeper Learning Curve.** Requires understanding Tensors and NN structure,

Feature	Scikit-learn (Sklearn)	TensorFlow (TF)
	`.predict()`). It requires minimal code and little knowledge of matrix algebra.	though the **Keras** high-level API significantly simplifies model building for beginners.
Community Support	**Excellent and Stable.** Highly relied upon in academic research, data science, and business for tabular/structured data modeling.	**Massive and Active.** Backed by Google, it has the largest community, rapid updates, and an extensive ecosystem (TensorBoard, TF Serving). It dominates the DL research and industry application space.